CYBERSECURITY PRACTICAL REPORT

FOOTPRINTING, OSINT AND NETWORK SCANNING

NetworkWalks Cybersecurity Internship | Batch B084 | Week 2

Intern Name	Musadiq Mehmood Qureshi
Internship	NetworkWalks Cybersecurity Internship
Batch	B084
Week	Week 2
Project	Footprinting, OSINT and Network Scanning
Modules Covered	PM1 – Footprinting and Reconnaissance PM2 – Google Hacking Database (GHDB) PM3 – Maltego OSINT PM4 – theHarvester PM5 – Zenmap Network Scanning
Nature of Work	Educational internship practical activities
Report Date	13 November 2026

Public version – prepared for publication in a public GitHub repository. Sensitive identifiers have been intentionally excluded.

1. Liability Disclaimer

This report documents educational practical activities completed as part of the Week 2 NetworkWalks Cybersecurity Internship. Its purpose is to record what the author learned about footprinting, open-source intelligence (OSINT) and host discovery. It is provided for learning and review purposes only.

The report does not describe a penetration test, a vulnerability assessment or any form of exploitation. No attempt was made to gain access to any system, account, device or data, and nothing in this report should be read as a claim that a vulnerability or security weakness was found.

This report makes no statement about the authorization status of any third-party system. Information-gathering and scanning techniques must only be used against systems that a person owns or is explicitly authorized to test. Misuse of these techniques may be unlawful. Search results, publicly visible information or the ability to reach a system never amount to permission to access it.

The author accepts no responsibility for any use of the information in this report by others. Sensitive technical identifiers have been deliberately left out so that the report is suitable for public publication. This privacy review reduces risk but cannot guarantee that a public document is entirely free of risk.

2. Introduction

Week 2 of the NetworkWalks Cybersecurity Internship focused on the early, information-gathering stages of security work. The practical modules covered **footprinting**, **OSINT** and **network discovery**, using tools commonly found in Kali Linux.

Footprinting is the process of collecting publicly available information about a target, such as domain registration data, DNS information, web technologies and protective controls. **OSINT** (open-source intelligence) extends this by gathering and correlating information from public sources and presenting the relationships between pieces of data. **Network discovery** identifies which devices are active on a network, which is the starting point for understanding what needs to be monitored and protected.

This report covers five modules (PM1 to PM5). PM1 used six tools against the domain networkwalks.com. PM2 explored the Google Hacking Database. PM3 used Maltego for OSINT. PM4 involved opening theHarvester and becoming familiar with its interface. PM5 involved Zenmap host discovery in a virtual lab. All steps are described exactly as performed, and the limits of each activity are stated.

3. Objectives

The learning objectives of the five modules were as follows:

- **PM1:** Understand how publicly available domain, DNS, web technology and firewall information can be gathered using WHOIS, WhatWeb, Nslookup, cURL, WAFW00F and DNSRecon.
- **PM2:** Understand how Google dorks, as catalogued in the Google Hacking Database, reveal what is publicly indexed and how this relates to possible exposure risks.
- **PM3:** Learn how Maltego supports OSINT investigations through entities, transforms and visual relationship mapping.
- **PM4:** Become familiar with the theHarvester application and its interface as an OSINT tool.
- **PM5:** Understand Ping Scan and host discovery with Zenmap, and practise troubleshooting a tool that failed to launch.
- Practise documenting technical work accurately and in a privacy-conscious way.

4. Tools Used

The table below lists each tool used, its purpose and the module in which it appeared.

Tool	Purpose	Module
WHOIS	Look up publicly available domain registration and name server information.	PM1
WhatWeb	Identify technologies used by a website, such as the content management system and plugins.	PM1
Nslookup	Resolve a domain name through DNS.	PM1
cURL (curl -I)	Retrieve and inspect HTTP response headers.	PM1
WAFW00F	Detect whether a web application firewall is protecting a website.	PM1
DNSRecon	Inspect DNS information such as NS, MX, SPF/TXT and SRV record categories.	PM1
Google Hacking Database (GHDB)	Browse a catalogue of Google search queries (dorks) used to understand publicly indexed information.	PM2
Maltego	Perform OSINT investigations and visualise relationships between entities using transforms.	PM3
theHarvester	OSINT tool for gathering publicly available information. Only opened and explored in this week.	PM4
Zenmap	Graphical interface for Nmap, used for Ping Scan and host discovery.	PM5
Kali Linux on Oracle VirtualBox	Virtual lab operating system used for the practical exercises, with a NAT network configuration.	PM1, PM3–PM5

5. Activities Performed

5.1 PM1 – Footprinting and Reconnaissance

The target domain for this exercise was **networkwalks.com**. Six tools were used to collect publicly available information.

WHOIS was used to inspect the publicly available registration information for the domain, including domain registration details and name server information.

WhatWeb was used to identify the technologies behind the website. It identified that the site uses WordPress together with a download-manager plugin. Exact version numbers are intentionally not published in this report.

Nslookup was used to perform DNS resolution and review how the domain resolves. The resolved address is intentionally not published.

cURL was run with the -I option to inspect the HTTP response headers, and the response information was reviewed. A WordPress REST API endpoint was observed. Header values are not reproduced here.

WAFW00F was used to check for a web application firewall. It detected **ModSecurity (SpiderLabs)**.

DNSRecon was used to inspect DNS information. The NS, MX, SPF/TXT and SRV record categories were reviewed. Exact record values are intentionally not published.

Identifying a technology, firewall or DNS record is information gathering only. None of these observations is treated as a vulnerability. Evidence is in Section 10.1.

5.2 PM2 – Google Hacking Database (GHDB)

The Google Hacking Database is a public catalogue of search queries, known as Google dorks, that use advanced search operators to find specific types of publicly indexed content. Security professionals use dorks to understand what an organization has exposed to search engines and to identify possible exposure risks.

The following activities were performed:

- Opened the Google Hacking Database.
- Explored camera-related Google dork searches.
- Reviewed camera-related search results.
- Performed a math/PDF search exercise.
- Captured screenshots of the exercises.

Important clarification: this was search-result research only. No camera was opened or accessed, no private device was accessed, and no camera was successfully discovered or connected to. Private camera addresses, device URLs and credentials are not included in this report. A search result does not give permission to access a device or website.

Evidence is in Section 10.2.

5.3 PM3 – Maltego OSINT

Maltego was installed and opened in Kali Linux. Maltego represents information as entities and uses transforms to find related data, which is displayed as a visual graph.

The following steps were performed:

- Opened the Maltego application in Kali Linux.
- Created a domain entity.
- Entered the target domain networkwalks.com.
- Right-clicked the domain entity.
- Explored the email-related transforms.
- Executed the selected email-related transforms.
- Reviewed the related results.

This exercise demonstrated how Maltego supports OSINT investigations and visual relationship discovery. Discovered email addresses are not published, the number of results is not stated, and ownership of any address was not verified. Evidence is in Section 10.3.

5.4 PM4 – theHarvester

theHarvester was opened in Kali Linux and its interface was viewed in order to become familiar with the tool. This module was limited to opening the application and interface familiarization.

No harvesting command was completed and no data collection took place for this module. No email addresses, subdomains, hostnames, IP addresses or scan results are reported. Evidence is in Section 10.4 and shows only that the application was opened.

5.5 PM5 – Zenmap Network Scanning

Windows troubleshooting. Zenmap was first opened on Windows, but it did not launch successfully and displayed the error 0xc000007b. Zenmap did not work on Windows. The exercise was therefore continued in Kali Linux.

Lab environment. The practical was done in Kali Linux running inside Oracle VirtualBox with a NAT network configuration. It was a virtual lab exercise. It was not a scan of the author's physical Windows network or home network.

Activity. Zenmap was opened in Kali Linux and a Ping Scan (host discovery) was run against the virtual machine's NAT subnet. Three live hosts were identified in the virtualized lab environment.

Concepts. A Ping Scan checks which addresses in a range respond, without scanning for open ports or services. Host discovery is the process of finding which devices on a network are active, and live-host identification is the result of that process. The educational purpose was to understand how an inventory of active devices is built, which is a foundation for network monitoring and later security analysis.

No open ports, running services or vulnerabilities were identified or reported, and this was not a penetration test or vulnerability assessment. IP addresses, MAC addresses and network identifiers are intentionally not published.

Evidence note. The screenshot of Zenmap opened is actual application evidence. The two images for live hosts and network topology in Section 10.5 are illustrative samples only, labelled as such, and are not screenshots of the author's actual scan.

6. Findings and Observations

The observations below summarize what was actually seen in each module. They are general observations, not vulnerability findings.

Module	Observations
PM1	Domain registration and name server information is publicly available through WHOIS. The website uses WordPress and a download-manager plugin. DNS resolution and DNS record categories (NS, MX, SPF/TXT, SRV) were reviewed. HTTP response headers were available, and a WordPress REST API endpoint was observed. A web application firewall, ModSecurity (SpiderLabs), was detected.
PM2	The Google Hacking Database shows how search operators can be used to find specific types of indexed content, including camera-related content. No camera or device was accessed.
PM3	Maltego can create a domain entity and run email-related transforms to show related information visually. Results are not published and ownership was not verified.
PM4	theHarvester opened successfully and its interface was viewed. No harvesting results exist for this module.
PM5	Zenmap did not launch on Windows (error 0xc000007b). In Kali Linux on VirtualBox (NAT), a Ping Scan found three live hosts in the virtual lab. No ports, services or vulnerabilities were assessed.

7. Risk Analysis / Impact

This section discusses general security implications of the types of information handled in this week's exercises. It does not claim that any breach or vulnerability was found.

Area	General security implication	General concern
Exposed technology information	Knowing the CMS or plugins a site uses can help an attacker look for known weaknesses if the software is not kept updated.	Low to Medium
Public DNS and registration data	DNS records and registration data are public by design, but they show an organization's infrastructure and mail setup and should be kept accurate and minimal.	Low
HTTP headers and API endpoints	Headers and endpoints can show information about the platform. Unnecessary detail should be limited.	Low
Public search indexing	Content indexed by search engines may expose information the owner did not intend to publish, which Google dorks can reveal.	Medium
OSINT results	Publicly gathered information such as email addresses can support social engineering or phishing if collected and misused.	Low to Medium
Network and host discovery	Knowing which devices are active is the basis for deeper testing. Unknown or unmanaged devices on a network may be a risk.	Low

The concern levels are a general perspective on each type of information and are not ratings of any specific system. The exercises involved information gathering and host discovery only, and a piece of information being visible does not by itself mean a vulnerability exists.

8. Recommendations

Based on the topics covered, the following defensive practices are recommended:

1. **Perform reconnaissance responsibly**
   Only run footprinting, OSINT and scanning against systems you own or are explicitly authorized to test, and keep records of that scope.
2. **Reduce unnecessary information exposure**
   Review what is visible in HTTP headers, public pages and DNS records, and remove details that are not needed.
3. **Keep software updated**
   Update the CMS, plugins and other web components regularly and remove those that are not used.
4. **Review publicly indexed content**
   Periodically search for your own organization using search-engine operators to find content that should not be public, and have it removed.
5. **Keep the firewall configured and monitored**
   Keep a web application firewall enabled, tuned and monitored.
6. **Limit public contact information**
   Reduce exposed email addresses where possible and train staff to recognize phishing.
7. **Monitor the network**
   Run authorized discovery on your own networks regularly to keep an inventory of active devices, and investigate any unexpected device.

8. **Protect privacy in documentation**
   Remove IP addresses, MAC addresses, credentials and personal paths from any report or screenshot before it is made public.

9. Conclusion

Week 2 provided practical experience with the information-gathering stages of security work. I used six footprinting tools, explored the Google Hacking Database, used Maltego for OSINT, opened theHarvester and ran host discovery with Zenmap in a Kali Linux lab. I learned that a lot can be understood about a target from public information, that this information should be handled responsibly, and that observations must be documented accurately without overstating them.

The completion status of each module is shown below.

Module	Status	Notes
PM1	Completed	Six tools used against the target domain. Information gathering only.
PM2	Completed	GHDB opened and camera-related dork results reviewed. Search-result research only; no device accessed.
PM3	Completed	Domain entity created and email-related transforms executed and reviewed. Results not published.
PM4	Interface familiarization only	theHarvester opened and interface viewed. No harvesting command completed.
PM5	Completed in Kali Linux lab	Zenmap did not launch on Windows (error 0xc000007b). Ping Scan host discovery done in the VirtualBox NAT lab. No ports or services assessed.

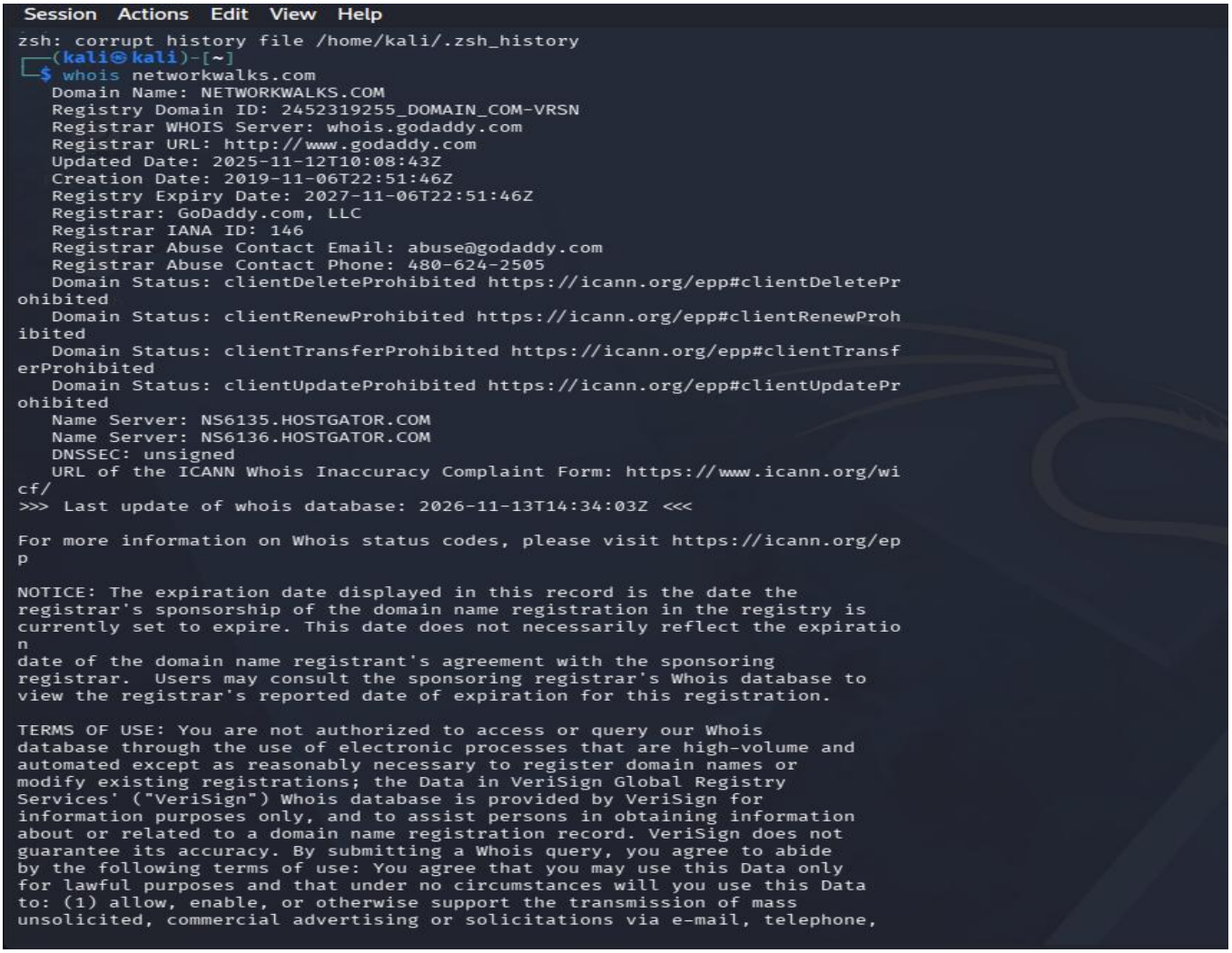
10. Evidence Collected

This section provides a dedicated page for each evidence screenshot. Each page reserves space for the image, the exact file name and a caption. The evidence is organised in folders PM1 to PM5.

Folder	File name	Type
PM1/	PM1_WHOIS.png	Screenshot evidence
PM1/	PM1_WhatWeb.png	Screenshot evidence
PM1/	PM1_WAFW00F.png	Screenshot evidence
PM2/	PM2_Task1_GHDB_Camera_Dorks.png	Screenshot evidence
PM2/	PM2_Task2_Math_PDF_Search.png	Screenshot evidence
PM2/	PM2_GHDB_Opened.png	Screenshot evidence
PM3/	PM3_Maltego_Homepage.png	Screenshot evidence
PM3/	PM3_NetworkWalks_Email_Transforms.png	Screenshot evidence
PM4/	PM4_theHarvester_Homepage.png	Screenshot evidence (application opened only)
PM5/	PM5_Live_Hosts.PNG	ILLUSTRATIVE SAMPLE – not actual scan evidence
PM5/	PM5_Network_Topology.PNG	ILLUSTRATIVE SAMPLE – not actual scan evidence
PM5/	PM5_Zenmap_Opened.PNG	Screenshot evidence (application opened)

10.1 PM1 – Footprinting and Reconnaissance Evidence
Folder: PM1/. The following pages are reserved for the WHOIS, WhatWeb and WAFW00F screenshots.

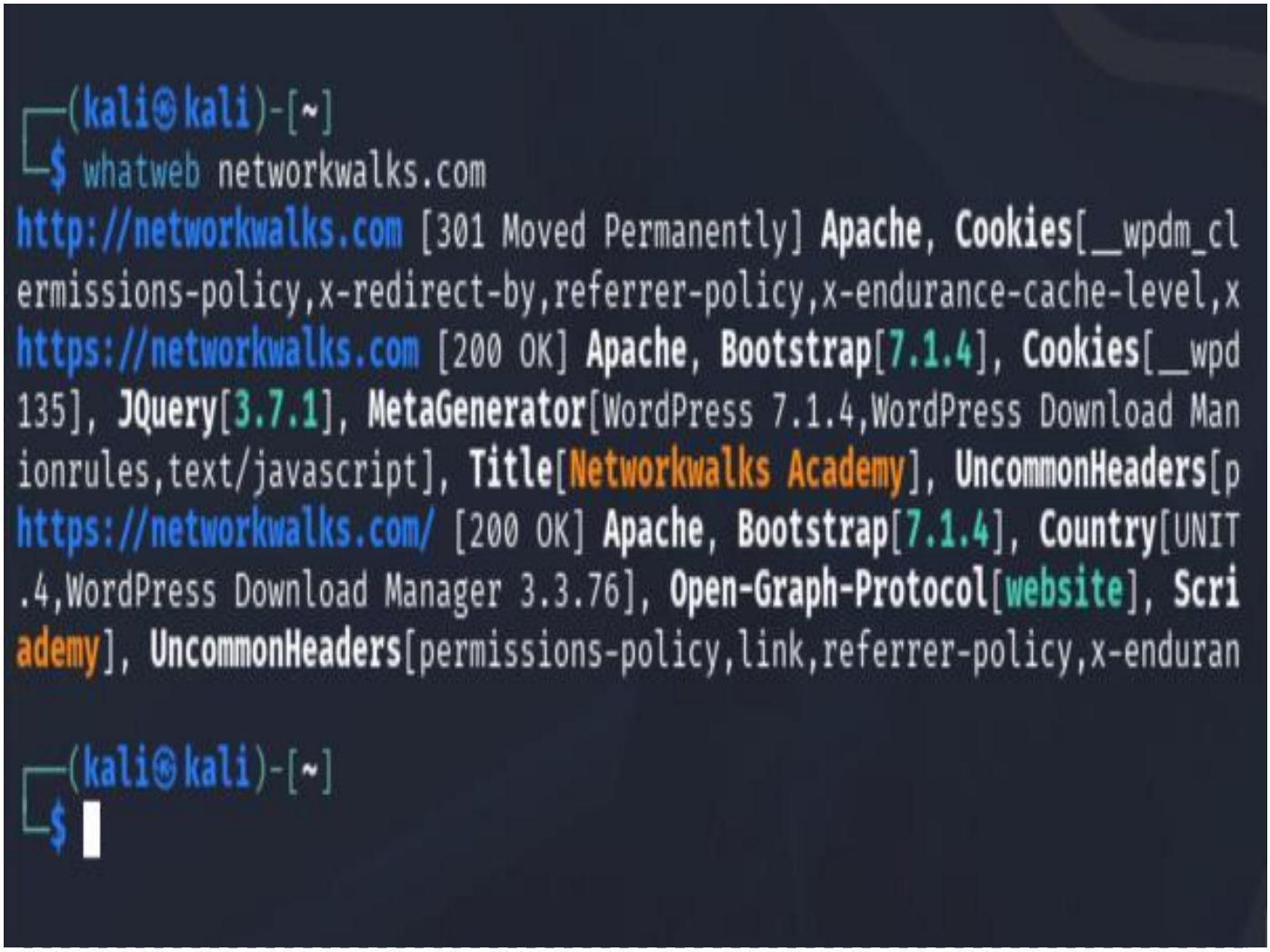
Evidence Screenshot: PM1_WHOIS.png



File: PM1/PM1_WHOIS.png

Caption: WHOIS domain registration information review.

Evidence Screenshot: PM1_WhatWeb.png



File: PM1/PM1_WhatWeb.png

Caption: WhatWeb website technology identification.

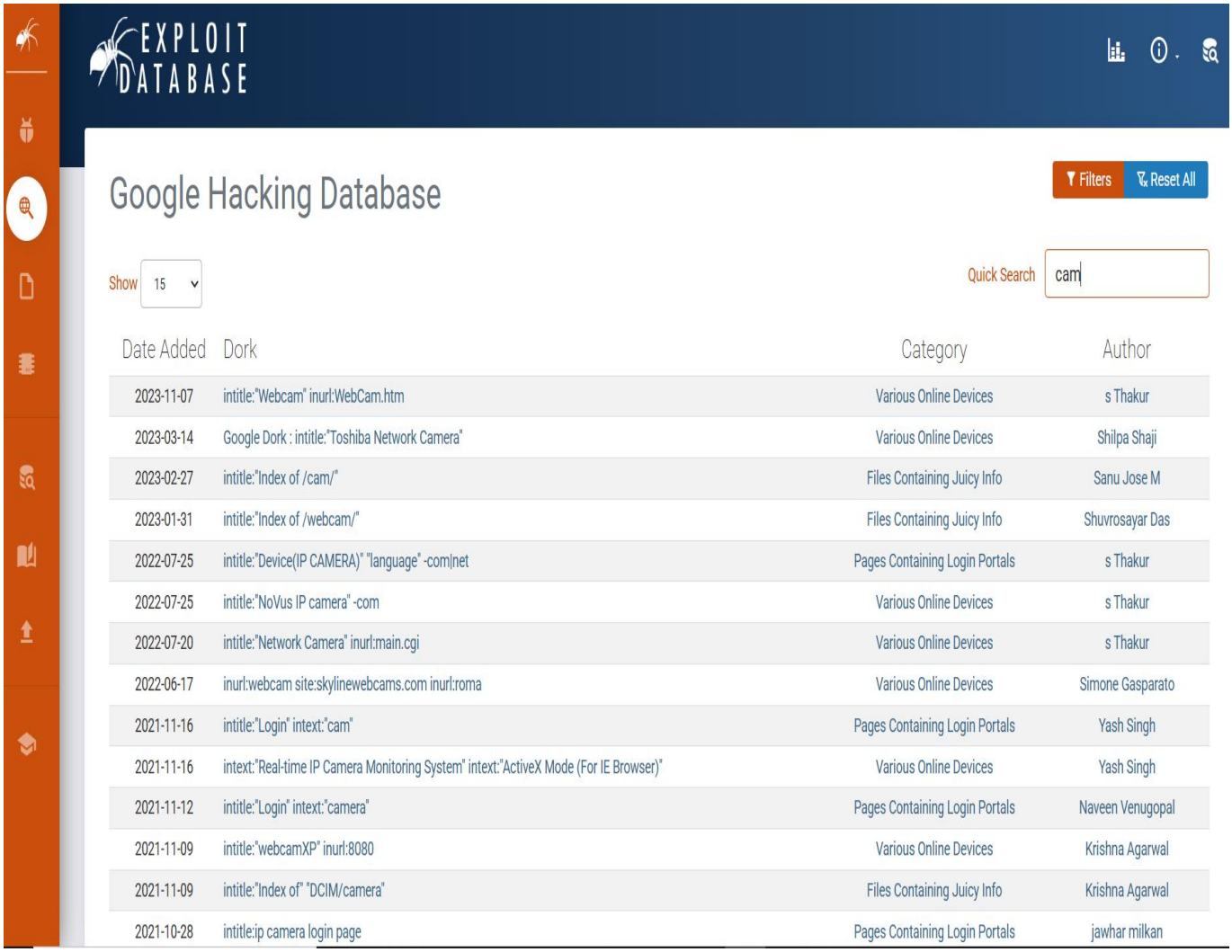
Evidence Screenshot: PM1_WAFW00F.png



10.2 PM2 – Google Hacking Database Evidence

Folder: PM2/. These screenshots show search-result research only; no camera or device was accessed.

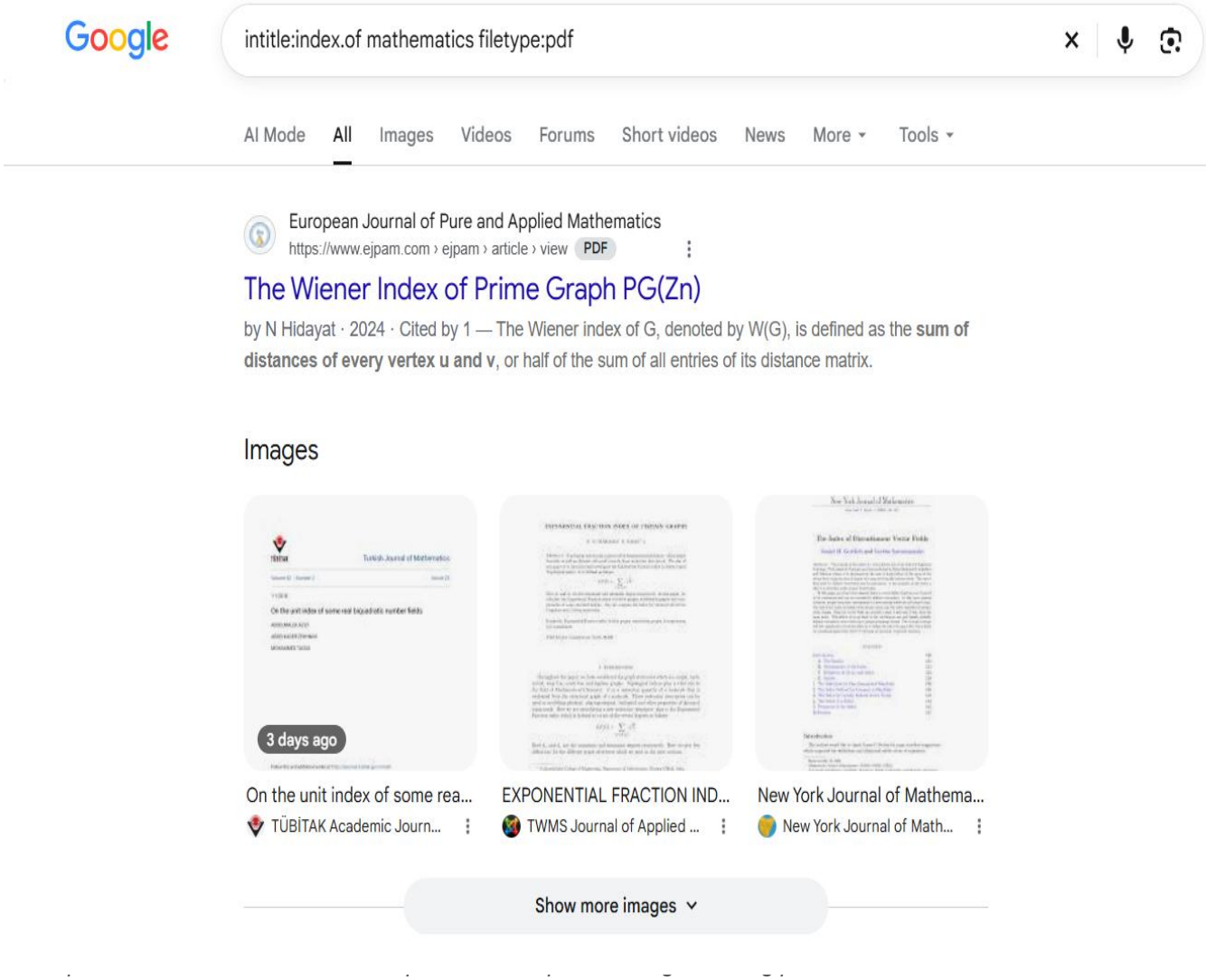
Evidence Screenshot: PM2_Task1_GHDB_Camera_Dorks.png



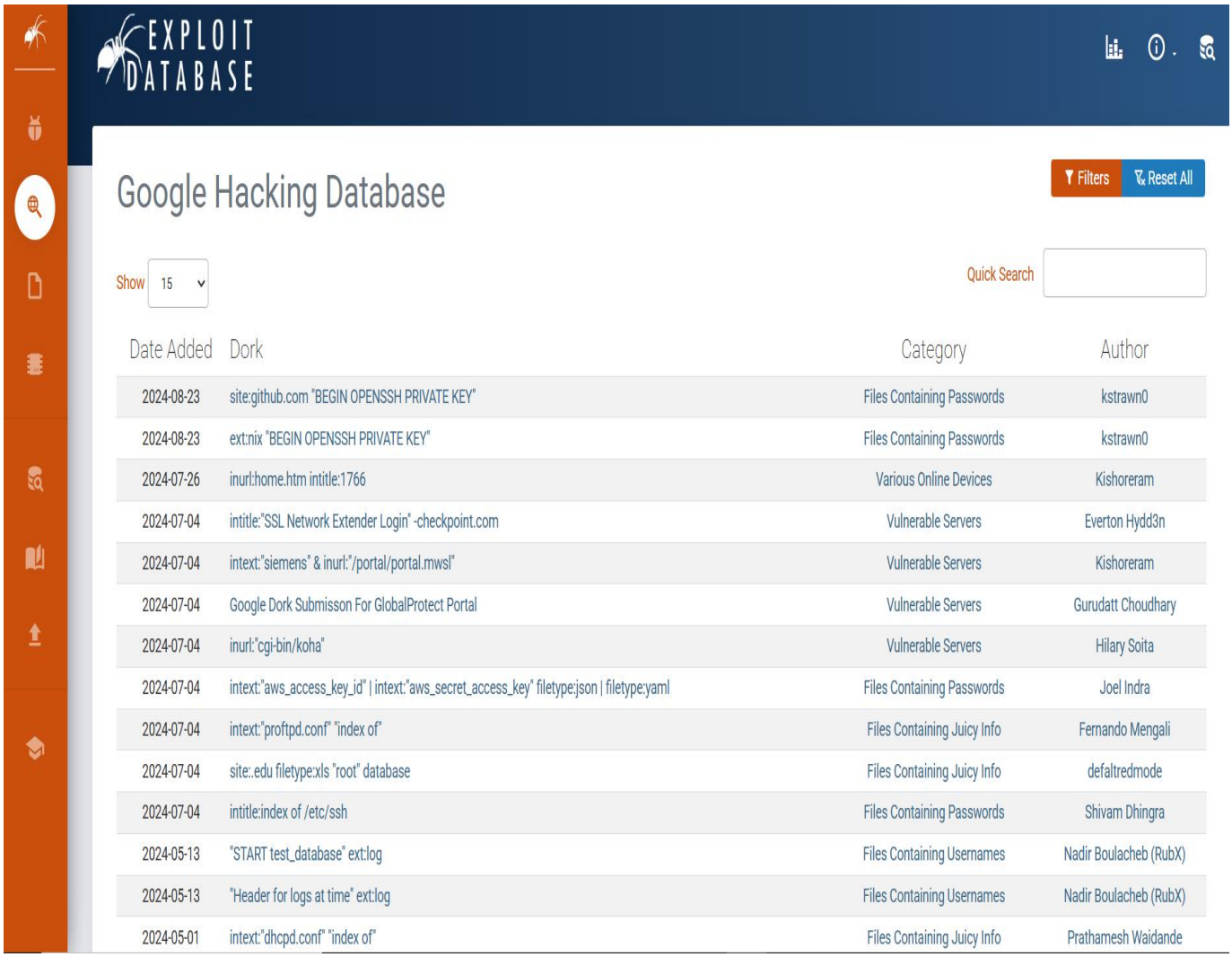
File: PM2/PM2_Task1_GHDB_Camera_Dorks.png

Caption: Camera-related Google dork exploration in the Google Hacking Database (search-result research only).

Evidence Screenshot: PM2_Task2_Math_PDF_Search.png



Evidence Screenshot: PM2_GHDB_Opened.png



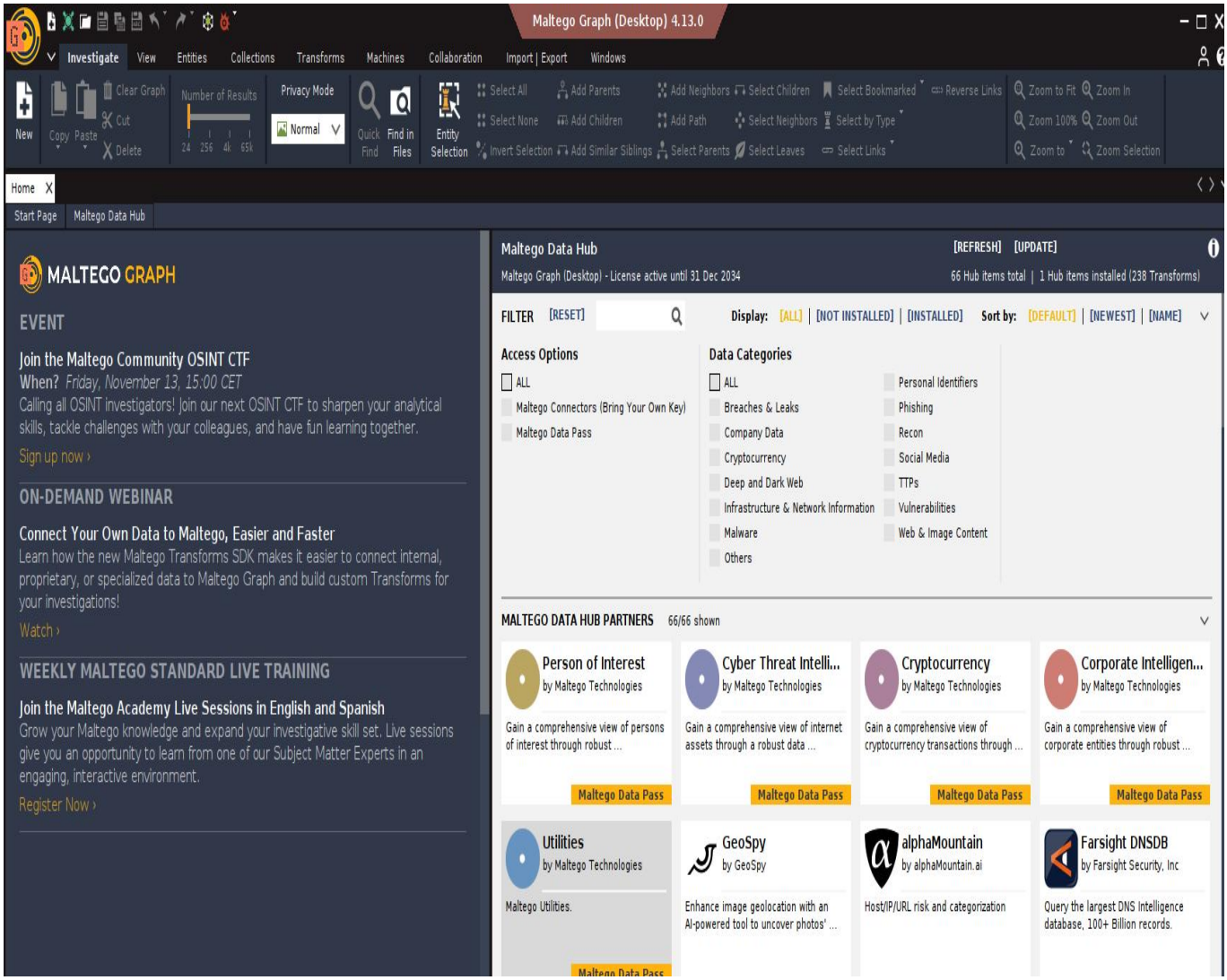
Date Added	Dork	Category	Author
2024-08-23	site:github.com "BEGIN OPENSSH PRIVATE KEY"	Files Containing Passwords	kstrawn0
2024-08-23	ext:nix "BEGIN OPENSSH PRIVATE KEY"	Files Containing Passwords	kstrawn0
2024-07-26	inurl:home.htm intitle:1766	Various Online Devices	Kishoreram
2024-07-04	intitle:"SSL Network Extender Login" -checkpoint.com	Vulnerable Servers	Everton Hydd3n
2024-07-04	intext:"siemens" & inurl:"/portal/portal.mwsl"	Vulnerable Servers	Kishoreram
2024-07-04	Google Dork Submisson For GlobalProtect Portal	Vulnerable Servers	Gurudatt Choudhary
2024-07-04	inurl:"cgi-bin/koha"	Vulnerable Servers	Hilary Soita
2024-07-04	intext:"aws_access_key_id" | intext:"aws_secret_access_key" filetype:json | filetype:yaml	Files Containing Passwords	Joel Indra
2024-07-04	intext:"proftpd.conf" "index of"	Files Containing Juicy Info	Fernando Mengali
2024-07-04	site:.edu filetype:xls "root" database	Files Containing Juicy Info	defaltredmode
2024-07-04	intitle:index of /etc/ssh	Files Containing Passwords	Shivam Dhingra
2024-05-13	"START test_database" ext:log	Files Containing Usernames	Nadir Boulacheb (RubX)
2024-05-13	"Header for logs at time" ext:log	Files Containing Usernames	Nadir Boulacheb (RubX)
2024-05-01	intext:"dhcpd.conf" "index of"	Files Containing Juicy Info	Prathamesh Waidande

File: PM2/PM2_GHDB_Opened.png

Caption: The Google Hacking Database opened in the browser.

10.3 PM3 – Maltego OSINT Evidence

Folder: PM3/. Discovered email addresses are not published in this report.



File: PM3/PM3_Maltego_Homepage.png

Caption: Maltego opened in Kali Linux.

Evidence Screenshot: PM3_NetworkWalks_Email_Transforms.png



File: PM3/PM3_NetworkWalks_Email_Transforms.png

Caption: Email-related transforms run on the domain entity.

10.4 PM4 – theHarvester Evidence

Folder: PM4/. The screenshot shows only that the application was opened.

Evidence Screenshot: PM4_theHarvester_Homepage.png



File: PM4/PM4_theHarvester_Homepage.png

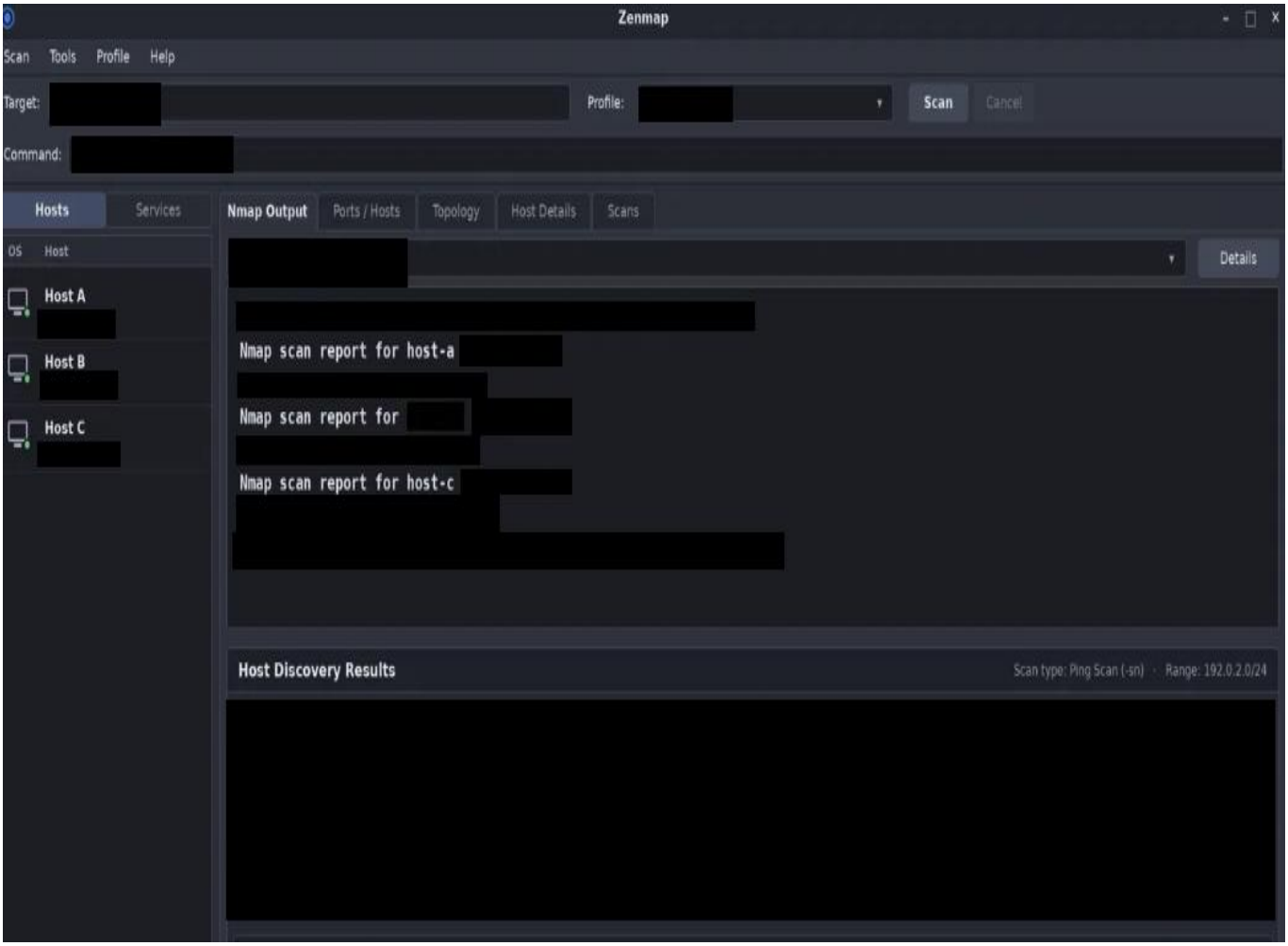
Caption: theHarvester opened in Kali Linux. This shows application launch only, not a completed harvesting command.

10.5 PM5 – Zenmap Network Scanning Evidence

Folder: PM5/. PM5_Zenmap_Opened.PNG is actual application evidence. PM5_Live_Hosts.PNG and PM5_Network_Topology.PNG are illustrative samples, clearly labelled, and must not be presented as genuine scan screenshots.

Evidence Screenshot: PM5_Live_Hosts.PNG

ILLUSTRATIVE SAMPLE – NOT ACTUAL SCAN EVIDENCE

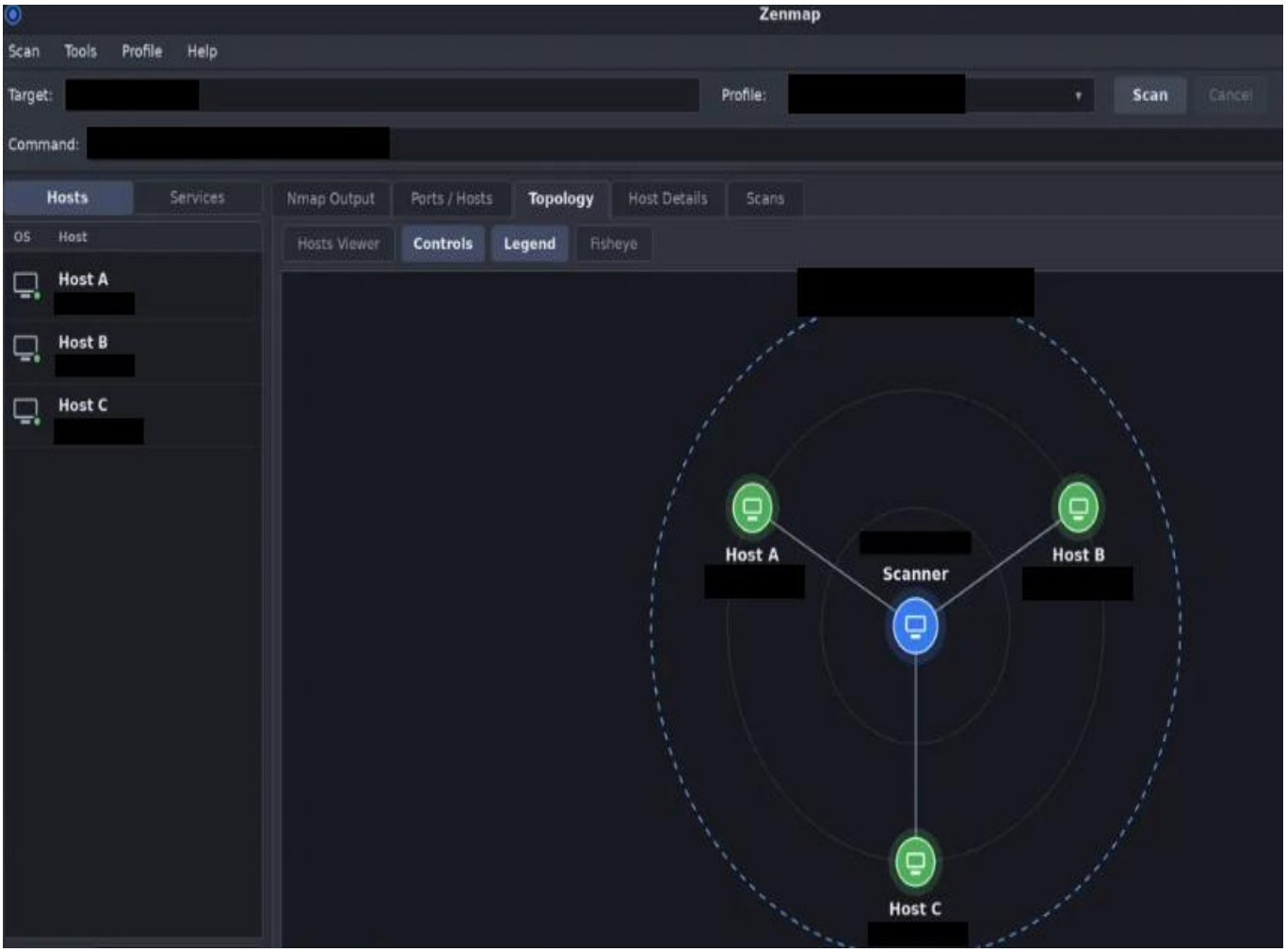


File: PM5/PM5_Live_Hosts.PNG

Caption: Illustrative sample of a live-host discovery view. This is not a screenshot of the actual scan.

Evidence Screenshot: PM5_Network_Topology.PNG

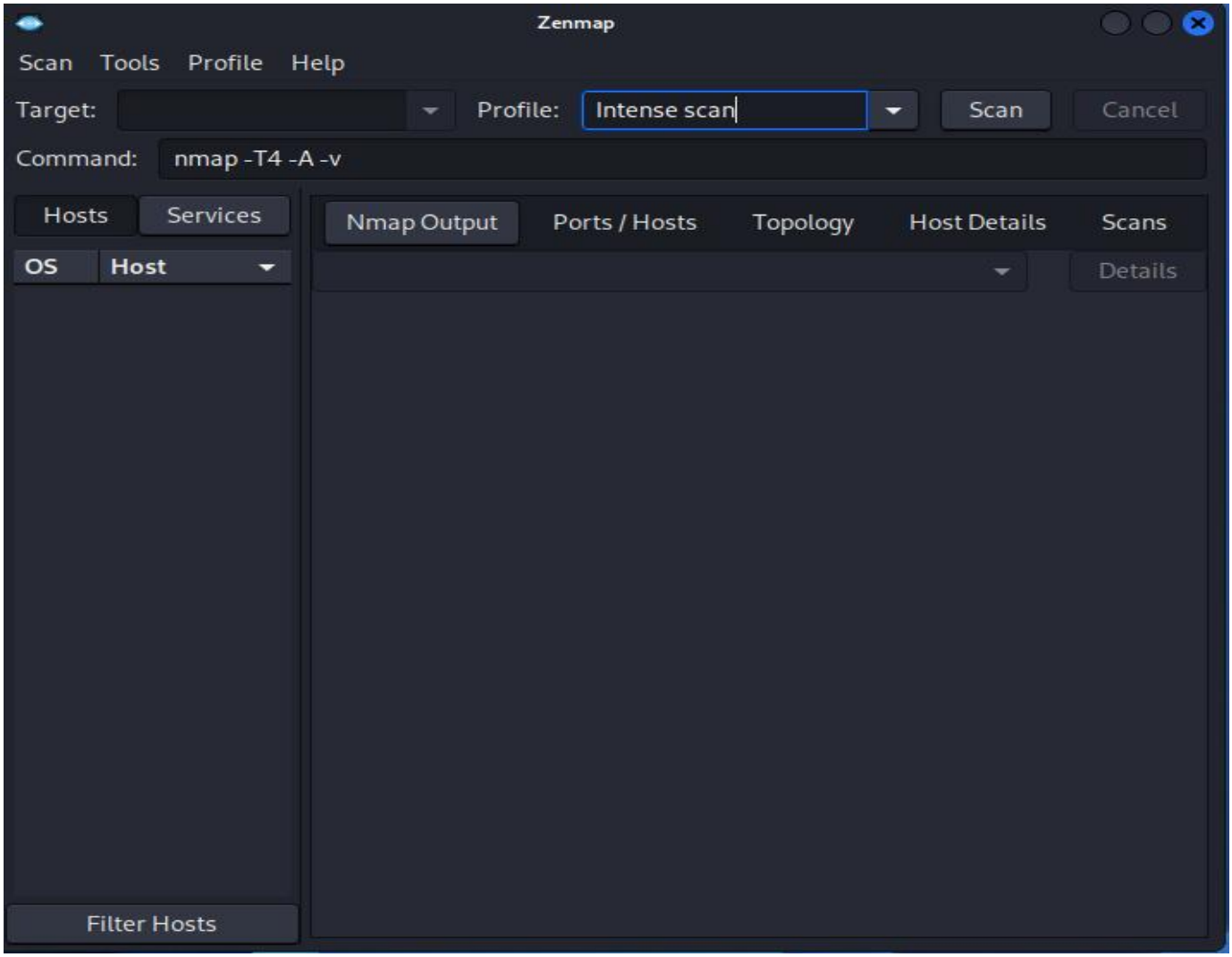
ILLUSTRATIVE SAMPLE – NOT ACTUAL SCAN EVIDENCE



File: PM5/PM5_Network_Topology.PNG

Caption: Illustrative sample of a network topology view. This is not a screenshot of the actual scan.

Evidence Screenshot: PM5_Zenmap_Opened.PNG



File: PM5/PM5_Zenmap_Opened.PNG

Caption: Zenmap opened in Kali Linux (actual application evidence).

11. Author and Internship Information

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